

5/12/18

Indian Institute of Engineering Science and Technology, Shibpur

Dual Degree (B.Tech-M.Tech) 3rd Semester Examination, 2018

Subject: Mathematics-III(MA-301)

(All Engineering Branches)

Use separate script for each half

Time : 3 hours

Full Marks : 70

First Half

Answer any eight questions

1. State axiomatic definition of probability. Using this definition show that for any two events A and B :

$$P(\overline{AB}) = P(B) - P(AB)$$

[5]

2. The chance that a doctor will diagnose a certain disease correctly is 60%. The chance that a patient will die by his treatment after correct diagnosis is 40% and the chance of death by wrong diagnosis is 70%. A patient of the doctor, who had the disease, dies. What is the probability that the disease was diagnosed correctly?

[5]

3. A man seeks advice regarding one of two possible courses of action from three advisers, who arrive at their recommendations independently. He follows the recommendations of the majority. The probabilities that the individual advisers are wrong are 0.1, 0.05 and 0.05 respectively. What is the probability that the man takes incorrect advice?

[5]

4. Find the constant k so that the function

$$f(x) = \begin{cases} kx^2, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

is a probability density function. Find (i) distribution function and (ii) $P(1 < X < 2)$.

[5]

5. Find the mean and variance of the Binomial distribution.

[5]

6. The length of bolts produced by a machine is normally distributed with mean 4 and standard deviation 0.5. A bolt is defective if its length does not lie in the interval (3.8, 4.3). Find the percentage of defective bolts produced by the machine. Given:

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0.6} e^{-t^2/2} dt = 0.7257, \quad \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0.4} e^{-t^2/2} dt = 0.6554.$$

[5]

7. (i) State and prove Tchebycheff's inequality for a continuous random variable.

(ii) Using Tchebycheff's inequality, find a lower bound for the probability of getting 64 to 184 driving licences issued by Road Transport Authority in a specific month. It is given that the number of driving licences issued per month be a random variable having mean $m = 124$ and standard deviation $\sigma = 7.5$. [3+2]

8. The probability density function a two-dimensional random variable (X, Y) is given by

$$f(x) = \begin{cases} k(x+y), & 0 < x+y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find k and evaluate $P(X < \frac{1}{2}, Y > \frac{1}{4})$. [5]

9. Fit a straight line to the following data:

x_i	y_i
1	2.4
2	3
3	3.6
4	4
6	5
8	6

[5]

10. Define correlation coefficient between two random variables and show that it lies between -1 and +1. [5]

11. In a partially destroyed Laboratory record of an analysis of correlation data, the following results only are legible: $\text{Var}(X)=9$.

Regression equations: $8X - 10Y + 66 = 0$, $40X - 18Y = 214$.

Find (i) the mean values of X and Y , (ii) $\rho(X, Y)$, (iii) σ_Y . [5]

12. If S^2 be the sample variance of a random sample $\{X_1, X_2, \dots, X_n\}$ of size n drawn from a population with mean μ and standard deviation σ , then prove that the statistic $s^2 = \frac{n}{n-1} S^2$ is an unbiased estimator of σ^2 . [5]

13. (i) Define consistent estimator for a population parameter.

(ii) Let $\{X_1, X_2, \dots, X_n\}$ be a random sample drawn from a population having a parameter θ and $T_n = T(X_1, X_2, \dots, X_n)$. If $E(T_n) \rightarrow \theta$ and $\text{Var}(T_n) \rightarrow 0$ as $n \rightarrow \infty$, then show that T_n is a consistent estimator of θ . [2+3]

14. Find the maximum likelihood estimate of p for a Binomial (N, p) population. [5]

SECOND HALFAnswer any **THREE (03)** questions15. a) If $f(t)$ is a periodic function with period T , then show that

$$\mathcal{L}[f(t); s] = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

Hence obtain the Laplace transform of the periodic function given by $f(t) = \frac{t}{T}$ of period T , $0 < t < T$.

b) State convolution theorem.

Apply convolution theorem to evaluate the inverse Laplace transform of

$$\frac{s}{(s^2 + a^2)^2}$$

[(3 + 2) + (1 + 4)]

16. a) Solve the following ordinary differential equation using Laplace transformation technique

$$\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 2x = te^{-t}$$

It is given that at $t = 0$, $x = 0$ and $\frac{dx}{dt} = 0$.

b) Evaluate the followings :

$$i) \mathcal{L} [t(3\sin 2t - 2\cos 2t); s] \quad , \quad ii) \mathcal{L}^{-1} \left[\frac{2s - 3}{s^2 - s - \frac{3}{4}}; t \right].$$

[5 + (2\frac{1}{2} + 2\frac{1}{2})]

17. a) Reduce the feasible solution $x_1 = 2, x_2 = 1, x_3 = 1$ of the system of equations

$$x_1 + 4x_2 - x_3 = 5$$

$$2x_1 + 3x_2 + x_3 = 8$$

to two basic feasible solutions.

b) Find the extreme points, if any, of the following sets :

$$i) X = \{ (x_1, x_2) : |x_1| \leq 1, |x_2| \leq 1 \}$$

$$ii) X = \{ (x_1, x_2) : x_1^2 + x_2^2 \leq 1; x_1 \geq 0, x_2 \geq 0 \}.$$

c) Evaluate the value of

$$\mathcal{L} \left[\frac{1 - \cos t}{t}; s \right].$$

[6 + 1 + 3]

18. a) Solve the following LPP by simplex method :

$$\begin{aligned} \text{Maximize } Z &= 10x_1 + x_2 + 2x_3 \\ \text{Subject to } x_1 + x_2 - 2x_3 &\leq 10 \\ 4x_1 + x_2 + x_3 &\leq 10 \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

b) Show that the set $X = \{\bar{x} : |\bar{x}| \leq 2\}$ is a convex set.

c) Check whether the following sets are convex or not :

i) $X = \{(x_1, x_2) : x_2^2 \geq 4x_1\}$

ii) $X = \{(x_1, x_2) : x_1x_2 \leq 4\}$

[5 + 3 + 2]

19. Solve the following LPP using Charne's Big M method :

$$\begin{aligned} \text{Maximize } Z &= 2x_1 + 3x_2 \\ \text{Subject to } x_1 + x_2 &\leq 8 \\ x_1 + 2x_2 &= 5 \\ 2x_1 + x_2 &\leq 8 \\ x_1, x_2 &\geq 0. \end{aligned}$$

OR

Use Two Phase method to solve the following LPP :

$$\begin{aligned} \text{Maximize } Z &= 3x_1 - x_2 \\ \text{Subject to } 2x_1 + x_2 &\geq 2 \\ x_1 + 3x_2 &\leq 2 \\ x_1 &\leq 4 \\ x_1, x_2 &\geq 0. \end{aligned}$$

[10]